

VI Semester B.E. Examination
(Electronics and Communication Engineering)
Automotive Electronics (22EECC305)

Duration: 3 hours**Max. Marks: 100**

Note: i) Answer any TWO full questions from UNIT-I, any TWO full questions from UNIT-II and any ONE full question from UNIT-III.

UNIT-I

- 1
 - a. Develop a fuel control strategy for varying operating modes for demonstrating their effects on efficiency, emission and drivability in conditions like cold start, idle and high load. (08marks)
 - b. Illustrate AGILE design methodology applied to embedded software development for electronic control unit. (06marks)
 - c. Explain the configuration and working principle of EGR actuator. (06marks)
- 2
 - a. Propose an optimal method for estimating mass air flow rate using alternative sensors when the MAF sensor malfunctions and justify its suitability in terms of real time adaptability, accuracy and computational efficiency. (08marks)
 - b. Suggest and explain the sensor used for closed loop operation of an engine control system to maintain the desired air/fuel ratio. (06marks)
 - c. Elaborate on different segments of automotive industry and also discuss about automotive supply chain. (06marks)
- 3
 - a. Develop a closed loop control strategy for an engine management system to handle spikes in cylinder pressure followed by sharp oscillations. (08marks)
 - b. Suggest and explain the suitable exhaust emission control device, which converts toxic gases and pollutants in exhaust gases from an internal combustion engine into non-toxic pollutants to comply with the exhaust emission regulations. (06marks)
 - c. Explain the working principles of strain gauge MAP sensor. (06marks)
- 4
 - a. What is ride and handling of the vehicle? Describe the suitable control system to manage ride and handling. (08marks)
 - b. Calculate nominal and maximum T_{HEADER} , T_{RESPONSE} and T_{FRAME} , if LIN protocol is operating at 10 kbps baud rate and reserved time is set to 30% for transmitting four bytes of data. (06marks)
 - c. Explain the differences between CAN and CAN FD protocol. (06marks)
- 5
 - a. The fast moving car is turning at the corner, if the vehicle is turning less/more than the driver's intention, suggest a suitable control system along with breaker circuit configuration to overcome this condition. (08marks)
 - b. Explain the following with respect to FlexRay communication protocol
 - i) Bus level
 - ii) Bus guardian
 - iii) Communication cycle
 (06marks)
 - c. Explain the frame format of MOST25 protocol. (06marks)
- 6
 - a. Design the control algorithm, which is suitable for adaptive control system to maintain a safe following distance from the car ahead. (08marks)
 - b. Explain the physical layer of automotive ethernet, which is responsible for (06marks)

encoding digital data into analog signals transmitting them over twisted pair of copper cables and ensuring reliable communication in the automotive vehicles.

- c. Design a suitable safety system, that prevents the wheels from locking, reducing the risk of skidding and reducing stopping distance in emergency braking or panic braking.

(06marks)

UNIT-III

- 7 a. Explain the SAE level of driving automation with an example.

(08marks)

- b. Explain ISO 26262 functional safety principles.

(06marks)

- c. Determine the ASIL value for an anti lock braking system in the concept phase.

(06marks)

- 8 a. Illustrate the need of on-board diagnostic system in modern vehicles.

(08marks)

- b. Explain the methodology to generate trouble codes and used to identify vehicle faults.

(06marks)

- c. Suggest and explain a diagnostic protocol, which request diagnostic services from vehicle ECU to read data, clear faults and reprogram ECU's.

(06marks)